Thus, the overall AT problem for improving the jailbreak robustness of the LLM p_θ is given as

$$\min_\theta \{\alpha \mathcal{L}_{\text{adv}}(\theta, M, D^{(h)}) + (1-\alpha)\mathcal{L}_{\text{utility}}(\theta, D^{(u)})\}, \tag{3}$$

where $\alpha \in [0,1]$ is a factor that balances between the adversarial and utility sub-losses. The idea behind such a loss design is that: (1) help LLM learn to respond harmlessly even when strong jailbreak prompts present (achieved via $\mathcal{L}_{\text{adv}}$), (2) retain the utility of LLM gained from pre-training (achieved via $\mathcal{L}_{\text{utility}}$). Intuitively, a larger adversarial suffix length m during AT will help the LLM gain robustness against jailbreak attacks with longer adversarial suffixes.

4 Theoretical evidence

This section establishes the theoretical foundation of how "short-length" AT can defend against "long-length" jailbreaking. Our analysis is based on the in-context learning (ICL) theory [69, 47, 4], and we will bridge the ICL theory and the LLM AT problem defined in Eq. (3) later (in Section 4.2). Here we first introduce the necessary notations to describe the problem. To avoid confusion, we note that **all notations in this section will only be used within this section and have no relevance to those in other sections** (*e.g.*, Section 3).

In-context learning (ICL). In the ICL theory, a *prompt* with length N related to a specific *task* indexed by τ is defined as $(x_{\tau,1}, y_{\tau,1}, \cdots, x_{\tau,N}, y_{\tau,N}, x_{\tau,q})$, where $x_{\tau,i} \in \mathbb{R}^d$ is the i-th in-context training sample, $y_{\tau,i} \in \mathbb{R}$ is the label for the i-th training sample, and $x_{\tau,q} \in \mathbb{R}^d$ is the in-context query sample. Then, the task-specific ICL input E_τ is defined as

$$E_\tau := \begin{pmatrix} x_{\tau,1} & \cdots & x_{\tau,N} & x_{\tau,q} \\ y_{\tau,1} & \cdots & y_{\tau,N} & 0 \end{pmatrix} \in \mathbb{R}^{(d+1)\times(N+1)}. \tag{4}$$

Given an ICL input E_τ of task τ, the goal of an ICL model is to make a prediction based on E_τ for the query sample $x_{\tau,q}$. Such an ICL model design aims to model the ability of real-world LLMs in making decisions based on prompting without updating model parameters.

Linear self-attention (LSA) models. LSA models are a kind of linear transformer that has been widely adopted in existing theoretical ICL studies. [2] empirically show that LSA models share similar properties with non-linear ones and thus are useful for understanding transformers. We follow [69] to study the following single-layer LSA model,

$$f_{\text{LSA},\theta}(E_\tau) := \left[E_\tau + W^V E_\tau \cdot \frac{E_\tau^\top W^{KQ} E_\tau}{N}\right] \in \mathbb{R}^{(d+1)\times(N+1)},$$

where $\theta := (W^V, W^{KQ})$ is the model parameter, $W^V \in \mathbb{R}^{(d+1)\times(d+1)}$ is the value weight matrix, $W^{KQ} \in \mathbb{R}^{(d+1)\times(d+1)}$ is a matrix merged from the key and query weight matrices of attention models, $E_\tau \in \mathbb{R}^{(d+1)\times(N+1)}$ is the task-specific ICL input, and N is the prompt length. The prediction $\hat{y}_{q,\theta}$ for the query sample $x_{\tau,q}$ is given by the right-bottom entry of the output matrix of the LSA model, *i.e.*, $\hat{y}_{q,\theta}(E_\tau) := f_{\text{LSA},\theta}(E_\tau)_{(d+1),(N+1)}$. We further follow [69] to denote that

$$W^\square = \begin{pmatrix} W_{11}^\square & w_{12}^\square \\ (w_{21}^\square)^\top & w_{22}^\square \end{pmatrix} \in \mathbb{R}^{(d+1)\times(d+1)},$$

where $\square \in \{V,\ KQ\}$, $W_{11}^\square \in \mathbb{R}^{d\times d}$, $w_{12}^\square, w_{21}^\square \in \mathbb{R}^{d\times 1}$ and $w_{22}^\square \in \mathbb{R}$. Under this setting, the model prediction $\hat{y}_{q,\theta}$ can be further simplified as follows,

$$\hat{y}_{q,\theta}(E_\tau) := f_{\text{LSA},\theta}(E_\tau)_{(d+1)\times(N+1)} = \begin{pmatrix}(w_{21}^V)^\top & w_{22}^V\end{pmatrix} \cdot \frac{E_\tau E_\tau^\top}{N} \cdot \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} \cdot x_{\tau,q}. \tag{5}$$

Other notations. For any $n \in \mathbb{N}^+$, we denote $[n] := \{1, \cdots, n\}$. For any $A \in \mathbb{R}^{n\times m}$, we denote $\|A\|_{2,\infty} := \max_{1\le i\le m} \|A_{i,:}\|_2$, $\|A\|_2$ be the operator norm, and $\|A\|_F$ be the Frobenius norm. For any $A \in \mathbb{R}^{n\times n}$, we denote $\text{Tr}(A) := \sum_{i=1}^n A_{i,i}$. We use standard big O notations $\mathcal{O}(\cdot)$ and $\Theta(\cdot)$.

4.1 Problem definition for adversarial ICL

We now formalize the AT problem in ICL with the previously introduced notations. We focus on the linear regression task and introduce a novel ICL "suffix" adversarial attack, where in-context adversarial points are appended to the end of ICL inputs, to analyze the robustness of LSA models.

In-context linear regression. For any task indexed by τ, we assume that there is a task weight $w_\tau \in \mathbb{R}^d$ drawn from $w_\tau \sim \mathcal{N}(0, I_d)$. Besides, for any in-context training point $x_{\tau,i}$ $(1 \le i \le N)$ and the query point $x_{\tau,q}$ (see Eq. (4)), we assume that they are drawn from $x_{\tau,i}, x_{\tau,q} \sim \mathcal{N}(0, \Lambda)$, where $\Lambda \in \mathbb{R}^{d\times d}$ is a positive-definite covariance matrix. Moreover, the ground-truth labels of training points $x_{\tau,i}$ and the query point $x_{\tau,q}$ are given by $y_{\tau,i} = w_\tau^\top x_{\tau,i}$ and $y_{\tau,q} = w_\tau^\top x_{\tau,q}$.

ICL suffix adversarial attack. Our novel adversarial attack against ICL models is launched via concatenating (clean) prompt embedding matrices with adversarial embedding suffixes. Specifically, for an ICL input E_τ of length N (see Eq. (4)), we will form its corresponding adversarial ICL input $E^{\text{adv}}_{\tau,M} \in \mathbb{R}^{(d+1)\times(N+M+1)}$ by concatenating E_τ with an adversarial suffix of length M as follows,

$$E^{\text{adv}}_{\tau,M} := \begin{pmatrix} \underbrace{\begin{pmatrix} X_\tau \\ Y_\tau \end{pmatrix}}_{\text{Training Data of Length } N} & \underbrace{\begin{pmatrix} X^{\text{sfx}}_\tau + \Delta_\tau \\ Y^{\text{sfx}}_\tau \end{pmatrix}}_{\text{Adversarial Suffix of Length } M} & \underbrace{\begin{pmatrix} x_{\tau,q} \\ 0 \end{pmatrix}}_{\text{Query Sample From } E_\tau} \end{pmatrix}, \tag{6}$$

where $X_\tau := (x_{\tau,1} \ \cdots \ x_{\tau,N}) \in \mathbb{R}^{d\times N}$ and $Y_\tau := (y_{\tau,1} \ \cdots \ y_{\tau,N}) \in \mathbb{R}^{1\times N}$ denote the N original training samples and labels, and $X^{\text{sfx}}_\tau := (x^{\text{sfx}}_{\tau,1} \ \cdots \ x^{\text{sfx}}_{\tau,M}) \in \mathbb{R}^{d\times M}$, $Y^{\text{sfx}}_\tau := (y^{\text{sfx}}_{\tau,1} \ \cdots \ y^{\text{sfx}}_{\tau,M}) \in \mathbb{R}^{1\times M}$, and $\Delta^{\text{sfx}}_\tau := (\delta_{\tau,1} \ \cdots \ \delta_{\tau,M}) \in \mathbb{R}^{d\times M}$ denote the new M clean suffix samples, clean suffix labels, and adversarial perturbations. The clean suffix samples X^{sfx}_τ and labels Y^{sfx}_τ here follow the same distribution as those in-context data in the embedding E_τ, *i.e.*, $x^{\text{sfx}}_{\tau,i} \sim \mathcal{N}(0, \Lambda)$ and $y^{\text{sfx}}_{\tau,i} = w_\tau^\top x^{\text{sfx}}_{\tau,i}$ hold for every $i \in [M]$. For the adversarial perturbation matrix Δ_τ, we require each perturbation $\delta_{\tau,i}$ is restricted within a ball-sphere as $\|\delta_{\tau,i}\|_2 \le \epsilon$, where $\epsilon > 0$ is the perturbation radius. This aims to simulate that in jailbreak attacks, and each adversarial token is searched within a token vocabulary set of limited size.

The goal of the ICL adversarial attack is to add an optimal suffix adversarial perturbation matrix Δ_τ to maximize the difference between the model prediction $\hat{y}_q(E^{\text{adv}}_\tau)$ based on the adversarial ICL input E^{adv}_τ and the ground-truth query label $y_{\tau,q}$. We adopt the squared loss to measure such a prediction difference, which thus leads to the robust generalization error for the model f^{LSA}_θ as

$$\mathcal{R}^{\text{adv}}(\theta, M) = \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty} \le \epsilon} \frac{1}{2} |\hat{y}_{q,\theta}(E^{\text{adv}}_{\tau,M}) - y_{\tau,q}|^2, \tag{7}$$

where M is the length of the adversarial suffix and the expectation $\mathbb{E}_\tau[\cdot]$ is calculated over the randomness of w_τ, X_τ, X^{sfx}_τ, and $x_{\tau,q}$. As we aim to understand how the adversarial prompt length in AT would affect the robustness of LLM, Eq. (7) will also only focus on how the adversarial suffix length M in ICL adversarial attacks would affect the robust generalization error $\mathcal{R}^{\text{adv}}(\theta, M)$.

Adversarial in-context learning. Following previous studies on minimax AT [32, 20, 43, 12, 58], here we adopt a minimax AT loss to train the LSA model. Concretely, we first use the aforementioned ICL adversarial attack to synthesize adversarial prompts and then update the LSA model based on these adversarial prompts to help the model gain robustness against them. We further assume that the adversarial suffix length is fixed during AT, which thus leads to the following ICL AT problem,

$$\min_\theta \mathcal{L}^{\text{adv}}(\theta) := \min_\theta \mathcal{R}^{\text{adv}}(\theta, M_{\text{train}}) = \min_\theta \left\{ \mathop{\mathbb{E}}_{\tau} \max_{\|\Delta_\tau^\top\|_{2,\infty} \le \epsilon} \frac{1}{2} |\hat{y}_{q,\theta}(E^{\text{adv}}_{\tau,M_{\text{train}}}) - y_{\tau,q}|^2 \right\}, \tag{8}$$

where $\mathcal{L}^{\text{adv}}(\theta) := \mathcal{R}^{\text{adv}}(\theta, M_{\text{train}})$ is the AT loss in ICL and $M_{\text{train}} \in \mathbb{N}^+$ is the fixed adversarial suffix length during AT. We will perform AT with continuous gradient flow, and further following [69] to make the following assumption on the LSA model parameter initialization.

Assumption 1 (c.f. Assumption 3 in [69])**.** *Let $\sigma > 0$ be a parameter and $\Theta \in \mathbb{R}^{d\times d}$ be any matrix satisfying $\|\Theta\Theta^\top\|_F = 1$ and $\Theta\Lambda \ne 0_{d\times d}$. We assume*

$$W^V(0) = \begin{pmatrix} 0_{d\times d} & 0_{d\times 1} \\ 0_{1\times d} & \sigma \end{pmatrix}, \quad W^{KQ}(0) = \begin{pmatrix} \sigma\Theta\Theta^\top & 0_{d\times 1} \\ 0_{1\times d} & 0 \end{pmatrix}.$$

Recall in Eq. (5), w^V_{12}, w^{KQ}_{12}, and w^{KQ}_{22} do not contribute to the model prediction $\hat{y}_{q,\theta}(\cdot)$. Assumption 1 thus directly sets them to be zero at initialization. To ensure symmetric initialization, it further sets $w^V_{21}(0)$ and $w^{KQ}_{21}(0)$ to zero. We will see how Assumption 1 helps simplify the analysis of ICL AT.

4.2 Bridging ICL AT and LLM AT

We now discuss similarities between the ICL AT problem defined in Eq. (8) and the LLM AT problem defined in Eq. (3) to help motivate why ICL AT can be a good artifact to theoretically study LLM AT.

Firstly, in-context inputs (*i.e.*, E_τ defined in Eq. (4)) for LSA models are similar to prompt inputs for real-world LLMs. If we replace each token in an LLM prompt with its *one-hot encoding* defined over the token vocabulary space, then these one-hot encodings would be similar to in-context samples x_i in Eq. (4) since both of them are now "feature vectors". Besides, we note that each in-context label y_i in Eq. (4) can be seen as the "next-token prediction label" in real-world LLMs. The main difference is that in LLMs, the i-th token in a prompt can be seen as the i-th input token and the $(i-1)$-th next-token prediction label simultaneously, while in LSA models, the i-th in-context input and the $(i-1)$-th in-context label are explicitly separated into two terms x_i and y_{i-1}.

Secondly, the search for adversarial in-context samples (see Eq. (6)) in the ICL suffix adversarial attack is similar to the search for adversarial tokens in suffix jailbreak attacks. We note that each adversarial token in jailbreak prompts can be seen as replacing the "padding token". Thereby, from the point of view of one-hot encoding, searching for an adversarial token can thus be seen as applying an ℓ_2-norm adversarial perturbation within a radius of $\sqrt{2}$ to transform the one-hot encoding of the padding token to that of the adversarial token. This process is the same as the search for adversarial in-context samples in the ICL suffix adversarial attack defined in Eq. (7), which would perturb each in-context suffix sample $x_{\tau,i}^{\text{sfx}}$ within an ℓ_2-norm ball-sphere under a given radius $\epsilon > 0$.

Thirdly, motivations behind ICL AT and LLM AT are also similar to each other. Both of the two AT problems aim to enhance models' robustness via training them on some synthesized adversarial inputs. The adversarial inputs syntheses in ICL AT and LLM AT are also similar, as both of them aim to make targeted models behave wrongly via manipulating suffixes of input prompts. The difference is that suffix jailbreak attacks are *targeted adversarial attacks* aimed at inducing LLMs to generate *specified* harmful content while our ICL attack is an *untargeted adversarial attack* aimed at reducing the utility of linear regression prediction made by LSA models.

4.3 Training dynamics of adversarial ICL

We now start to analyze the training dynamics of the minimax ICL AT problem formalized in Eq. (8). The main technical challenge is that to solve the inner maximization problem in Eq. (8), one needs to analyze the optimization of the adversarial perturbation matrix Δ_τ. However, the matrix Δ_τ along with the clean data embedding E_τ and the clean adversarial suffix $(X_\tau^{\text{sfx}}, Y_\tau^{\text{sfx}})$ are entangled together within the adversarial ICL input $E_{\tau,M_{\text{train}}}^{\text{adv}}$, which makes it very difficult to solve the inner maximization problem and further analyze the ICL AT dynamics.

To tackle such a challenge, we propose to instead study the dynamics of a *closed-form upper bound* of the original AT loss $\mathcal{L}^{\text{adv}}(\theta)$. Formally, we will analyze the following surrogate AT problem:

$$\min_\theta \tilde{\mathcal{L}}^{\text{adv}}(\theta) := \min_\theta \Big\{ \sum_{i=1}^4 \ell_i(\theta) \Big\}, \tag{9}$$

where $\tilde{\mathcal{L}}^{\text{adv}}(\theta) := \sum_{i=1}^4 \ell_i(\theta)$ is the surrogate AT loss, $E_{\tau,M_{\text{train}}}^{\text{clean}} := \begin{pmatrix} X_\tau & X_\tau^{\text{sfx}} & x_{\tau,q} \\ Y_\tau & Y_\tau^{\text{sfx}} & 0 \end{pmatrix}$, and

$$\ell_1(\theta) = 2 \mathop{\mathbb{E}}_\tau \Big[((w_{21}^V)^\top \;\; w_{22}^V) \frac{E_{\tau,M_{\text{train}}}^{\text{clean}} E_{\tau,M_{\text{train}}}^{\text{clean}\top}}{N + M_{\text{train}}} \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} x_{\tau,q} - y_{\tau,q} \Big]^2,$$

$$\ell_2(\theta) = \frac{2\epsilon^4 M_{\text{train}}^2}{(N + M_{\text{train}})^2} \|w_{21}^V\|_2^2 \mathop{\mathbb{E}}_\tau \Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2 \Big],$$

$$\ell_3(\theta) = \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2} \mathop{\mathbb{E}}_\tau \Big[\|W_{11}^{KQ} x_{\tau,q}\|_2^2 \cdot \|((w_{21}^V)^\top \;\; w_{22}^V) \begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix} \|_2^2 \Big],$$

$$\ell_4(\theta) = \frac{2\epsilon^2 M_{\text{train}}}{(N + M_{\text{train}})^2} \|w_{21}^V\|_2^2 \cdot \mathop{\mathbb{E}}_\tau \Big[\| \begin{pmatrix} X_\tau^{\text{sfx}} \\ Y_\tau^{\text{sfx}} \end{pmatrix}^\top \begin{pmatrix} W_{11}^{KQ} \\ (w_{21}^{KQ})^\top \end{pmatrix} x_{\tau,q} \|_2^2 \Big].$$

The surrogate AT loss $\tilde{\mathcal{L}}^{\text{adv}}(\theta)$ in Eq. (9) is the closed-form upper bound for the original AT loss $\mathcal{L}^{\text{adv}}(\theta)$ in Eq. (8), as illustrated in the below Proposition 1 (see Appendix A.2 for the proof).